Low energy Hamiltonian $H = H_K + H_{K'}$

$$H_K = \sum_q{}' \left(\psi^+_{KAS}, \psi^+_{KBS}\right) \hbar v \begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix} \begin{pmatrix} \psi_{KAS} \\ \psi_{KBS} \end{pmatrix}$$

($\sum_q{}'$ denotes sum over $\vec{q}$ centered around $\vec{K}$)

$$= \sum_q{}' \psi^+_{KS} v \vec{\sigma} \cdot \vec{p} \, \psi_{KS} \qquad (\hbar \vec{q} = \vec{p})$$

and $\psi_{KS} = \begin{pmatrix} \psi_{KAS} \\ \psi_{KBS} \end{pmatrix}$, $\psi^+_{KS} = \begin{pmatrix} \psi_{KAS} \\ \psi_{KBS} \end{pmatrix}^+ = \left(\psi^+_{KAS}, \psi^+_{KBS}\right)$

see p.89

$$H_{K'} = \sum_q{}' \left(\psi^+_{K'BS}, \psi^+_{K'AS}\right) (-\hbar v) \begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix} \begin{pmatrix} \psi_{K'BS} \\ \psi_{K'AS} \end{pmatrix}$$

$$= \sum_q{}' \psi^+_{K'S} (-v) \vec{\sigma} \cdot \vec{p} \, \psi_{K'S}$$

To study optical properties, we need to think how external field couples to electrons

$\vec{p} \to \vec{p} - \frac{e}{c} \vec{A}$ (see p.64 for usual electrons)

Consider uniform electric field $\vec{E}(t) = -\frac{1}{c} \frac{\partial \vec{A}}{\partial t}$

Vector potential $\vec{A}(t) = \vec{A} e^{-i\omega t}$

$\Rightarrow \vec{E}(t) = \frac{i\omega}{c} \vec{A}(t)$

Note that physical electric field $\vec{E}_{ph}(t) = \mathrm{Re}(\vec{E} e^{-i\omega t}) = \frac{1}{2}\left(\vec{E} e^{-i\omega t} + \vec{E}^* e^{i\omega t}\right)$

We are interested in optical (vertical) transitions and focus now on H_k (analysis for $H_{k'}$ is the same)

$$\Rightarrow \quad H_k \to \sum_p{}' \; \psi^+_{ks} \; v\,\vec{\sigma}\cdot\left(\vec{p} - \frac{e}{c}\vec{A}\right) \psi_{ks}$$

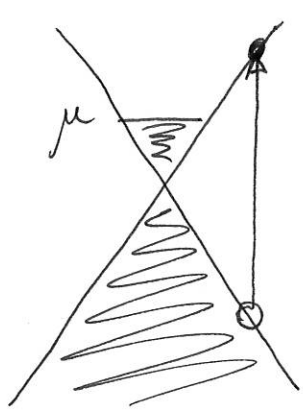
$$= \underbrace{H_k^{(0)}}_{\substack{\text{unperturbed} \\ (\vec{A}=0)\text{ Hamiltonian}}} + \underbrace{H_k^{(int)}}_{\text{interaction with light}}$$

$$H^{(int)} = \sum_p{}' \; \psi^+_{ks} \left(-\frac{ev}{c}\,\vec{\sigma}\cdot\vec{A} \right) \psi_{ks}$$

$$= \sum_p{}' \; \psi^+_{ks} \left(-\frac{ev}{c}\,\frac{c}{2i\omega}\,\vec{\sigma}\cdot\vec{E} \right) \psi_{ks}$$

should be $\vec{E}_{ph} = \vec{E}\cos(\omega t)$

Interband transition



Initial state:
electron with momentum $\vec{p}$ and energy $E_-(\vec{p}) = -vp$

Final state:
electron with $\vec{p}$ and energy $E_+(\vec{p}) = +vp$

$$\hat{H}_0 |i\rangle = -vp\,|i\rangle$$

Thus $$v\begin{pmatrix} 0 & p_x - ip_y \\ p_x + ip_y & 0 \end{pmatrix}\begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix} = \underbrace{-vp}_{E_-}\begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix}$$

Use $p_x = p\cos\varphi$, $p_y = p\sin\varphi$

$$vp\begin{pmatrix} 0 & e^{-i\varphi_p} \\ e^{i\varphi_p} & 0 \end{pmatrix}\begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix} = vp\begin{pmatrix} e^{-i\varphi_p}\psi_B \\ e^{i\varphi_p}\psi_A \end{pmatrix} = -vp\begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix}$$

$$\Rightarrow \psi_A = -e^{-i\varphi_p}\psi_B$$

Normalize: $|\psi_A|^2 + |\psi_B|^2 = 1 \Rightarrow |i\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} -e^{-i\varphi_p} \\ 1 \end{pmatrix}$

Final state: same $\vec{p}$, but $E = E_+ = vp$

$$\Rightarrow e^{-i\varphi_p}\psi_B = +\psi_A$$

$$\Rightarrow |f\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} e^{-i\varphi_p} \\ 1 \end{pmatrix}$$

$$\Rightarrow \langle f|H'|i\rangle = \frac{1}{\sqrt{2}}\left(e^{i\varphi_p}, 1\right)\frac{iev}{2\omega}\begin{pmatrix} 0 & E_x - iE_y \\ E_x + iE_y & 0 \end{pmatrix}\begin{pmatrix} -e^{-i\varphi_p} \\ 1 \end{pmatrix}\frac{1}{\sqrt{2}}$$

$$= \frac{1}{2}\frac{iev}{2\omega}\left(e^{i\varphi}, 1\right)\begin{pmatrix} +E_x - iE_y \\ -(E_x + iE_y)e^{-i\varphi} \end{pmatrix} =$$

$$= \frac{1}{2}\frac{iev}{2\omega}\left\{(E_x - iE_y)e^{i\varphi} - (E_x + iE_y)e^{-i\varphi}\right\}$$

$$= \frac{iev}{2\omega}\left\{iE_x\sin\varphi - iE_y\cos\varphi\right\}$$

Transition rate

$$\int \frac{d\vec{p}}{(2\pi\hbar)^2} \frac{2\pi}{\hbar} \left|\langle f | H' | i \rangle\right|^2 \delta(E_f - E_i - \hbar\omega) =$$

$$= \int \frac{dp \cdot p}{(2\pi\hbar)^2} \int_0^{2\pi} d\varphi \; \frac{2\pi}{\hbar} \left(\frac{ev}{2\omega}\right)^2 \left(E_x \sin\varphi - E_y \cos\varphi\right)^2 \delta(2vp - \hbar\omega)$$

$$= \frac{2\pi}{\hbar^3} \left(\frac{ev}{2\omega}\right)^2 \frac{1}{2}\left(E_x^2 + E_y^2\right) \underbrace{\int \frac{dp \cdot p}{2\pi} \delta(2vp - \hbar\omega)}_{\frac{1}{2\pi} \cdot \frac{1}{2v} \cdot \frac{\hbar\omega}{2v}}$$

$$= \left(\frac{ev}{2\omega}\right)^2 \frac{\omega E^2}{8 v^2 \hbar^2} = \frac{e^2 E^2}{4 \cdot 8\, \omega \hbar^2}$$

~~Joule Heating~~

Absorbed energy $= \text{rate} \cdot \hbar\omega = \dfrac{e^2 E^2}{4 \cdot 8 \hbar}$

Needs to be multiplied

by $4 = \underset{\text{spin}}{2} \cdot \underset{\text{valley}}{2} \quad \Rightarrow \quad \dfrac{e^2 E^2}{8\hbar}$

This is the energy of Joule heating $= \frac{1}{2} \sigma E^2$

$$\Rightarrow \sigma = \frac{e^2}{4\hbar} = \frac{1}{8} \frac{2e^2}{\hbar} \simeq \frac{1}{8} \frac{1}{13000\,\Omega}$$

Incident energy flux $= \frac{c}{4\pi} \frac{1}{2}|E|^2$

$$\Rightarrow \frac{\frac{e^2 E^2}{8\hbar}}{\frac{c E^2}{8\pi}} = \frac{\pi e^2}{\hbar c} = \frac{\pi}{137} \approx 2.3\%$$